ELEMENTS AND COMPOUNDS

Definitions

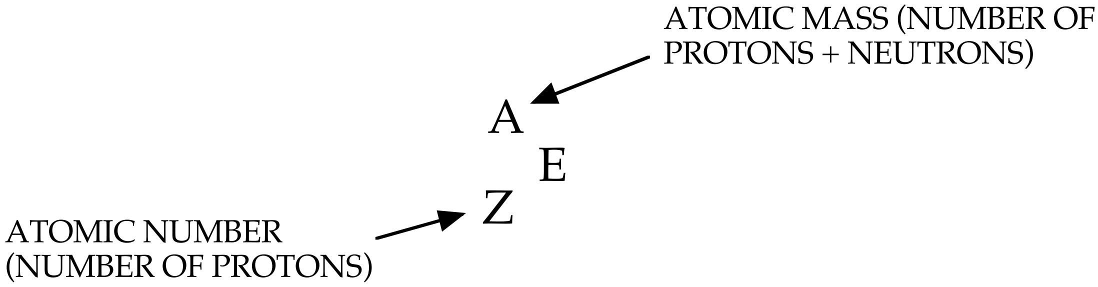
Element: substance that contains atoms that are all the same.
(Elements have atoms with the same number of protons in the nucleus.)

Atomic Number: number of protons in the nucleus.

Mass Number: number of protons and neutrons in the nucleus.

Each element has its own particular number of protons, neutrons and electrons.

These numbers are represented using the following general symbol.



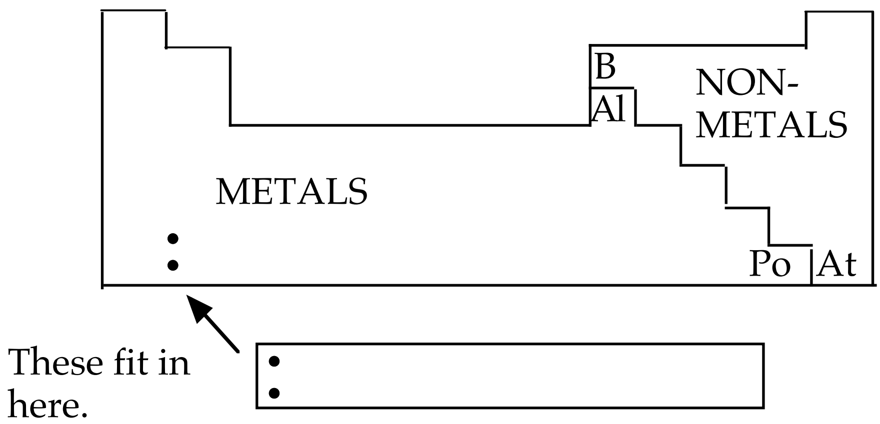
Examples

ELEMENT	PROTONS	NEUTRONS	ELECTRONS
${}^{24}_{12}Mg$	12	12	12
${}^{23}_{11}Na$	11	12	11
${}^{238}_{92}U$	92	146	92

There are 92 naturally-occurring elements and 22 man-made ones.

These are arranged by atomic number in the Periodic Table.

The elements can be classified as metals and non-metals.



Metals are found on the left hand side of the table, non-metals to the right.

Along the diagonal shown are ***metalloids*** or ***semi-metals*** that show properties of both metals and non-metals.

The symbols for elements always start with a capital letter, and the second letter (if used) is always lower case.

e.g. Al - aluminium C - carbon

The symbol is based on the English name, though some are based on Latin, Arabic and German names.

NON-ENGLISH		**ENGLISH**	
sodium	Na	hydrogen	H
potassium	K	helium	He
iron	Fe	lithium	Li
copper	Cu	beryllium	Be
silver	Ag	boron	B
tungsten	W	carbon	C
gold	Au	nitrogen	N
mercury	Hg	oxygen	O
lead	Pb	fluorine	F
tin	Sn	neon	Ne
		magnesium	Mg
cobalt	Co	aluminium	Al
manganese	Mn	silicon	Si
bromine	Br	phosphorus	P
iodine	I	sulfur	S
zinc	Zn	chlorine	Cl
nickel	Ni	argon	Ar
barium	Ba	calcium	Ca